Parker Ferrer

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Education

University of Georgia, Morehead Honors College — GPA: 3.8/4.0
M.S. Mechanical Engineering (Non-Thesis) – Accelerated BS/MS Program — May 2028
B.S. in Mechanical Engineering, **B.S.** Electrical & Electronics Engineering – Double Major — December 2027
Foundation Fellowship Class of 2028: Premier UGA academic scholarship (25/∼16,000 students)

Technical Skills

Software: CAD (SolidWorks [*CSWA Certified*], OnShape, Creo, Inventor, AutoCAD, Fusion 360), Fusion 360 CAM, KiCAD
Tools: CNC Mill/Router/Lathe, Manual Mill/Lathe, 3D Printer, Waterjet, Laser Cutter, Saws (Table, Miter, Band)
Programming: Python, Java, C++ (Arduino IDE), MATLAB

Work Experience

Caterpillar — Peoria, IL
Systems Integration Intern — Autonomy and Automation Division — *May 2026 – August 2026*

- Designed custom mechanical mounting brackets and ruggedized power distribution enclosures for autonomous test vehicles in Creo, leveraging FDM prototyping to resolve sub-3mm clearance constraints.
- Executed root-cause failure analysis and hardware diagnostics on vehicle-integrated autonomous systems, driving targeted structural modifications that increased subsystem reliability and reduced field repair downtime by 35%.
- Developed high-current wiring harness layouts and enclosure mounting interfaces with cross-functional autonomy teams, reducing electro-mechanical interference and reducing technician maintenance downtime by 60%.

ERMCO Power Partners — Athens, GA
Continuous Improvement Intern — *May 2025 – August 2025*

- Architected and labeled a 13,000-component ground-truth dataset for automated computer vision inspection, evaluating error modes with vendor engineers to improve model detection accuracy to 97%.
- Engineered and prototyped 3 iterations of passive alignment fixtures in OnShape, constraining transformer rotational alignment to within 3° across 95% of units for automated optical inspection.
- Optimized 2,000-lb capacity coil oven transport carts utilizing V-groove rail systems, releasing detailed CAD manufacturing drawings that reduced structural deflection by 15%.

AutomationDirect.com — Cumming, GA
Engineering Design Intern — *June 2023 – August 2024*

- Upgraded and validated 2 industrial automation testbeds integrating PLCs, HMIs, and electrical actuators to demonstrate real-time closed-loop control algorithms under 50ms latency.
- Spearheaded mechanical design for a novel $4m^3$ industrial additive manufacturing system, engineering the structural frame, multi-axis Cartesian stepper gantry, segmented thermal build plate, and modular enclosure.
- Engineered complete multi-voltage electrical infrastructure (220V AC / 48V DC / 24V DC), authoring full schematics and integrating high-current stepper drivers, closed-loop thermal safety relays, and sensor communication networks.

Projects & Leadership

PedalSwap — Athens, GA
Founder | Lead Product Design Engineer — *November 2025 – Present*

- Architected compact electromechanical packaging for a modular audio system, executing 12 rapid 3D-printing iterations via OnShape to convert breadboard electronics into an award-winning prototype (2nd place at Georgia Tech Idea to Prototype).
- Designed a toolless, blind-mate magnetic coupling mechanism for modular electronics, optimizing contact alignment and user experience to earn 2nd place at the Georgia Tech InVenture Prize ($10,000 prize and patent funding)
- Executed DFMA optimizations for mass production, drafting utility patent drawings, and vetting component suppliers as part of Georgia Tech's CREATE-X accelerator to commercialize hardware.

UGA Robotics Club — Athens, GA
Multi-Project Chief Engineer | Club Treasurer — *June 2024 – Present*

- Managed a 15-member interdisciplinary team through CAD design and CNC machining of an autonomous robot, elevating national ranking from 30th to 8th out of 45 teams at the IEEE Hardware Competition.
- Tightened mechanical subsystem tolerances to 0.005in and optimized sensor mounting, reducing control software compute overhead by 20%.
- Instituted formal Stage-Gate CAD review protocols and sub-assembly milestone tracking, eliminating interdisciplinary integration errors and achieving 100% on-time hardware fabrication.

UGA Cultivate Lab — Athens, GA
Undergraduate Researcher — Adaptive Instrumentation Hardware Development — *September 2024 – August 2025*

- Developed and fabricated a 4-iteration mechatronic instrument fixture over 4 major iterations within a 3-week timeline, engineering custom servo-driven end-effectors to achieve high-frequency strumming kinematics with 5ms tolerance.
- Co-authored and presented peer-reviewed research at the ASEE 2025 Annual Conference:
"Bridging the Gap: Autoethnographic Insights into Project-Based Learning in Electrical Engineering".